



Pedestrian safety perception and urban street settings

Yunmi Park & Max Garcia

To cite this article: Yunmi Park & Max Garcia (2020) Pedestrian safety perception and urban street settings, International Journal of Sustainable Transportation, 14:11, 860-871, DOI: [10.1080/15568318.2019.1641577](https://doi.org/10.1080/15568318.2019.1641577)

To link to this article: <https://doi.org/10.1080/15568318.2019.1641577>



Published online: 30 Jul 2019.



Submit your article to this journal [↗](#)



Article views: 6168



View related articles [↗](#)



View Crossmark data [↗](#)



Citing articles: 84 View citing articles [↗](#)



Pedestrian safety perception and urban street settings

Yunmi Park^a  and Max Garcia^b

^aArchitectural and Urban Engineering Systems, Ewha Womans University, Seoul, Korea (The Republic of); ^bEnvironmental Design, Auburn University College of Architecture Design and Construction, Auburn, Alabama, USA

ABSTRACT

Design and construction professionals are entrusted to design built environments that can assure public safety. Crime statistics is one indicator that reflects the correlation between built environments and public safety. Acknowledging the perception of pedestrians, specifically their psychological and emotional reactions toward certain public space settings would be another effective parameter to develop a good foundation to create more livable and safer neighborhoods. With these in mind, the analyses in this study have attempted to explore the relationship between street conditions and the perception of pedestrian safety in Auburn Alabama, USA. The research design primarily consists of an online survey that obtained information about (1) respondents' brief personal characteristics, (2) their walking times, (3) their levels of anxiety regarding certain street settings based on daytime and nighttime photographs (closed-ended questions on a Likert scale from 1 to 5), and (4) their reasoning behind these ratings (open-ended). The results show that proper street lighting is the main contributor to enhancing the feeling of safety on streets followed by the morphologic complexity of streets formed by various business types, outdoor dining, street performers, and other pedestrian activities. Other than physical settings, the study also found that age and gender influenced the level of anxiety among pedestrians.

ARTICLE HISTORY

Received 15 October 2018
Revised 4 July 2019
Accepted 5 July 2019

KEYWORDS

Online-survey; pedestrian;
safety perception; street
design; street setting

1. Introduction

Public urban streets regularly facilitate various pedestrian events such as conversations, daily activities, leisure walks, and political protests; they function not only as channels for movement but also as important public spaces within a city. Therefore, interest in the creation of lively streets has grown and planning recommendations are often codified using Complete Street guidelines, pedestrian master plans, and downtown revitalization plans. In addition, scholarly research supports endeavors to explore the consequences of vibrant street settings on human behavior, psychological health, and satisfaction regarding social interaction (Duany, Plater-Zyberk, Krieger, & Lennertz, 1991; Talen, 1999), physical activity (Brownson, Baker, Housemann, Brennan, & Bacak, 2001; Cohen et al., 2007; Lee & Moudon, 2004), and the alleviation of environmental degradation associated with vehicles (Frumkin, 2002). In addition, the impact of street environments on perception, primarily regarding the fear of crime, has been investigated frequently. Although previous studies have been thorough, additional empirical evidence about the popular perception of street settings and its impact on willingness to walk would be valuable for future decisions in urban planning and design.

This study asked two major questions, particularly to young adults who are prone to walking or using public transportation more often than members of the older generations: (1) Do people perceive certain types of street settings

as more uncomfortable or safe? (2) Which street settings most affect this perception? According to Sivak and Schoettle (2016), a significant drop in the prevalence of driver's licenses—almost 16% between 1983 and 2014—exists among the younger generation. Nicks (2016) also indicated that members of the younger generation prefer not to own vehicles because of the high maintenance costs (32%), the ease of obtaining transit from other sources (31%), biking or walking (22%), public transportation (17%), and concern for the environment (9%). This phenomenon partially recalls the changing trends that result from the recent interest in new urbanist planning strategies (e.g., compact cities, new urbanism, urban villages, and smart growth) and the efforts of public education and outreach. The changing attitudes of young adults also show the potential for creating active and livable urban streets since they seem willing to spend more time on the streets and less time inside private vehicles. Thus, this research can help clarify the needs and perceptions of today's young people and thus guide future urban design choices, policies, and master plans that aim to create viable streets for pedestrians.

2. Literature review: Urban street settings and people's safety perceptions

Researchers and practitioners in planning have offered various suggestions for creating vibrant, safe, and livable streets.

Ewing and Clemente (2013) recommended five dimensions for measuring the quality of urban settings: imageability, human scale, transparency, complexity, and safety and tidiness. Imageability refers to a community's context and is understood as the number of publicly recognizable establishments such as historic buildings, place signs, and identifiers. The human scale is specific to the design of street features that have been built for human comfort such as building height and setback or street furniture. Transparency relates to the visibility, shadows, and hiding spots, which could be evaluated by the presence of first-floor windows, active use on the streets, or occupied storefronts. Complexity accounts for the occurrence and variety of activities along streets that induce human movement, which could be measured using business density and types, pedestrian activity, or outdoor dining. Safety and tidiness are commonly desired traits in the public arena; this refers to the quality of sidewalks, related pedestrian infrastructure, and the presence of litter and graffiti. Mehta (2007) used the following design elements to categorize street design: the variety of businesses on a block, independent stores, permeability at the street front, personalization and signs on the street front, the number of community gathering places, the articulation of building façades, seating provided by public authorities, and commercial seating provided by businesses, all of which highlight the interface designs that smoothly connect private and public spaces and mixed-use. Similarly, Park (2015) emphasized the importance of design principles that can help connect buildings and streets effectively and jointly by investigating the potential of Christopher Alexander's pattern language to create more vibrant downtown spaces. Activity pockets, arcades, canvas roofs, sitting walls, street cafés, and positive outdoor spaces are highlighted as primary elements of this mode of thought. Montgomery (1998) recommended improving cities' activities, image, and form through different design elements such as development intensity, mixed-use, fine grain, human scale, city blocks, permeability, contact, visibility, the horizontal grain of streets, public realm, movement, green spaces, water spaces, landmarks, visual stimulation, attention to detail, and architectural style. Even these principles were suggested to help "make a city," most of these suggestions directly encourage people to interact on streets and actively participate in street life, thus increasing pedestrian traffic (Grant, 2002; Jacobs, 1961).

Beyond this discussion, a growing body of planning research has narrowed the scope of this topic to street safety and security. Based on Jacobs (1961)'s ideas about revitalized street life and increased "natural surveillance" using first-floor windows, which in turn support Newman's (1976) "defensible space theory" and crime prevention through environmental design (CPTED), it is widely believed and empirically supported that a built environment influences both crime and the fear of crime (Lee, Park & Jung, 2016; Minnery & Lim, 2005; Montoya, Junger & Ongena, 2016; Perkins, Meeks & Taylor, 1992). CPTED is comprised of detailed design elements intended to reduce both crime and the fear of crime in the community through increased

surveillance, access control, and territorial enforcement (Cozens, Saville & Hillier, 2005). Natural surveillance can occur as a result of high levels of street activity and population density caused by well-mixed land use, windows overlooking sidewalks and parking lots, street designs that attract pedestrian and bicycle traffic, and the proper placement of street lighting (Poyner & Webb, 1997). Natural territorial reinforcement with planned activities in public areas may attract more people while also strengthening the perception that these areas are well maintained. Fear of crime, as opposed to actual criminal activity, has also attracted attention in urban planning. Many "no-go" areas are designated as such because of the fear of crime rather than crime itself (Hunter & Baumer, 1982); the risk in these areas tends to be fairly low (Painter, 1996). The reduction of fear of crime through design eventually also mediates actual crime thanks to commonly used urban streets that feature high levels of pedestrian traffic. Where does this fear of crime or anxiety in public spaces originate? Previous research suggests that personal characteristics and experiences function as the strongest determinants (Cozens et al., 2005). The literature has found that female residents tend to feel more vulnerable than male residents, particularly at night, as women generally experience greater concerns about defending against offenders (Fisher, 1995; Fox, Nobles & Piquero, 2009; Hale, 1996; Ortega & Myles, 1987). Through interviews, Fanghanel (2015) found that women are attached to certain safekeeping practices because they depend on full knowledge of public spaces. Contemporary social life and the subjectification of women have resulted in a deep desire to reduce the fear of public spaces, even in places that they only know from secondhand accounts (Fanghanel, 2015). Other demographic attributes such as age and race/ethnicity have often been discussed. Elderly residents tend to express higher levels of fear than their younger counterparts (Ferraro & Grange, 1987; Fetschenhauer & Buunk, 2005; Gibson, Zhao, Lovrich, & Gaffney, 2002). The race/ethnicity factor produces a broad variety of mixed outcomes. Some researchers suggest that nonwhite residents—that is, black (Skogan & Maxfield, 1981) and Hispanic (Taylor, Eitle & Russell, 2009)—report higher levels of fear than their white counterparts (Fox et al., 2009). Other reports found no significant differences based on race/ethnicity (Chiricos, Hogan & Gertz, 1997; Fisher, 1995). People from different cultural backgrounds tend to express various levels of fear regarding similar settings. Ratnayake (2016) concluded that some international students in Australia—Asian females (28%) and Asian males (23%)—expressed slightly higher levels of fear than domestic Australian students (11%); the paper did not clearly state the reasons for this, but they may stem from the varying physical settings of public spaces and the lower levels of familiarity with the subject area and culture.

Another frequently mentioned factor is the physical setting. People are often reluctant to venture into neighborhoods that appear deteriorated or socially disordered (Hale, 1996; Painter, 1996). Incivilities, particularly "the actual presence of disorder-related cues," are also positively related with the perception of crime-related problems (Perkins

et al., 1992). Painter (1996) pointed out that after-dark hours cause the most anxiety because poor visibility—low levels of visual access—can result in blind spots or shadows in which criminal activities could take place. Thus, improving street lighting can be a cost-effective means to reduce the fear of crime (Hale, 1996). Fisher and Nasar (1992) introduced the “prospect-refuge-escape” concept based on Appleton (1975)’s “prospect-refuge theory.” According to them, safe places limit hiding spots for offenders and provide pedestrians with high visibility and multiple escape routes. Mixed land use with various functions can also help mediate the fear of crime by providing “natural watchers and guardians in sufficient numbers” (Jacobs, 1961) and promoting social integration among community members, thereby establishing informal social control (Maxfield, 1984).

The previous literature has successfully started identifying the determinants of safety perception (e.g., gender, age, ethnicity, and physical settings) and the rationales behind anxiety on the streets. The development of measures of the quality of streets, which are still mostly qualitative, can help investigate the relationship between street conditions and safety perception. However, a small gap remains that this paper can fill. As mentioned above, younger peoples’ awareness about walking and using non-motorized vehicles has changed dramatically due to cost burdens, planning strategies, and public education. Additional empirical evidence on safety perceptions of certain urban street design features could encourage increases in pedestrian activity among young adults, who have the potential to change the private-vehicle-oriented culture in the future. In addition, several previous studies have investigated safety perception through scaled rating surveys; these suggest new methods to map pedestrian activities by interviewing survey respondents and added a sentiment analysis that explores pedestrians’ perception of street settings in greater depth using open-end questions.

3. Research design

3.1. Study area

The city of Auburn, Alabama, USA, is home to Auburn University and is a principal city of the Auburn–Opelika metropolitan area. In 2015, about 28,000 people were enrolled at Auburn University, accounting for almost 44% of the city’s population. The study area is close to Auburn University’s campus and was chosen because this area is downtown of Auburn; the streets in this area account for many of the city’s pedestrians. In addition, most of the students (survey respondents) were familiar with the area due to its proximity to the main campus entrance. The study area included the main street, called Magnolia Avenue (1 mile or 1.6 km long) that intersects with South College Street to form a major downtown juncture. The west side of the intersection is mainly comprised of student housing, restaurants, bars, and gift shops. Commercial land uses are directly adjacent to the east side of the intersection with residential complexes and single-family housing units situated further east (Figure 1).

Perception of specific physical street settings, based on photographs of areas A, B, and C were addressed through open-ended questions that requested explanations for each rating. Area A is located on West Magnolia Avenue and functions as a transition zone between commercial and residential neighborhoods. Part of the front lot is used for parking and there is a large building setback on the lot. Area A overlooks three different campus entrances and is equipped with a variety of pedestrian facilities such as traffic islands, sidewalks, and crosswalks. Area B is a downtown commercial area with street parking lots at the buildings’ frontage. Area C is located on East Magnolia Avenue and is primarily a residential area.

The quality of the urban setting in Areas A, B, and C was rated by the authors using modified evaluation items based on Ewing and Clemente (2013) and Bereitschaft (2017)’s urban form quality measures as shown in Table 1. The ratings were compared and discussed until consensus was reached between authors. Quality scoring helped assess the quality of certain street settings and analyze the correlations in the surveyed pedestrian safety perception. The ratings followed a scale of good (two points), moderate (one point), or poor (zero points) and were then averaged. Quality scoring did not necessarily impose quantitative values with exact intervals between good, moderate, or poor, but it did highlight broad characteristics among the areas for a clearer understanding of the street settings. Overall, Area B possessed the highest level of quality (total score of 1.37), followed by Area C (1.05) and then Area A (0.68); the score for each area indicates that their quality is distinguishable. Area A earned a high score in terms of safety and tidiness but had low scores for imageability, human scale, and complexity categories. Area B is a downtown area and thus received the highest score among the three areas. Area C is primarily a residential area with high levels of tidiness but low levels of morphologic complexity and transparency. Like Area A, Area C relies heavily on tidiness, but its morphologic complexity score was higher than that of Area A. This accounts for mixed land use, which contributes to a street façade with variety and visibility. Based on the quality ratings and major land uses, areas A, B, and C were chosen for the photographic survey questionnaires.

3.2. Survey questions and procedures

With Institutional Review Board (IRB) exemption approval, a survey was distributed online via Qualtrics from February 25th to March 4th, 2018. One hundred and ninety-four respondents answered the survey, but only 160 (about 83%) were used for the analyses because 34 respondents either did not complete the survey or exited at an early stage. The analysis only included Auburn University students.

Sixteen survey questions were constructed in four categories:

- a. General participant characteristics (five questions): Auburn University student status, age, gender, race, and

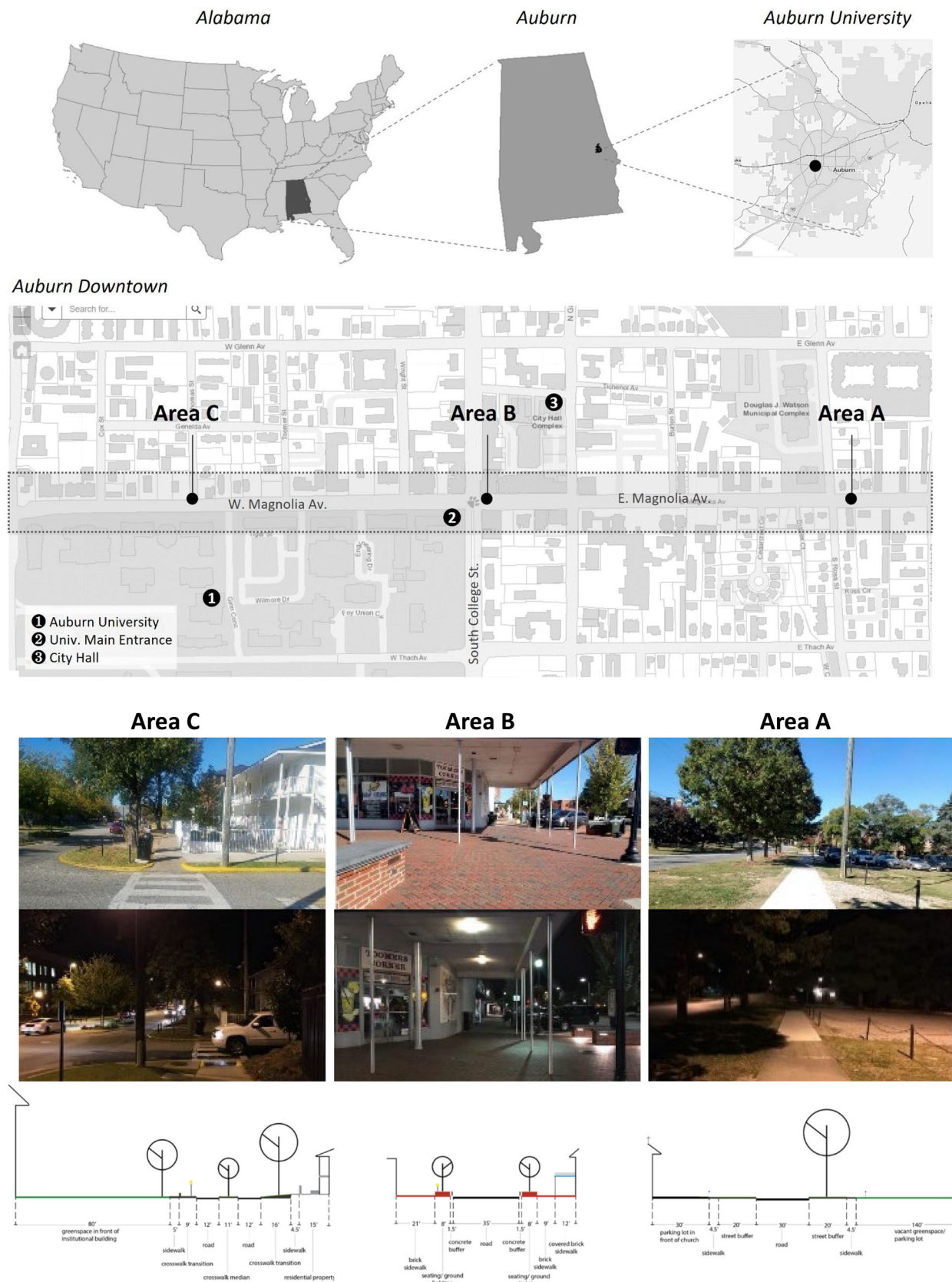


Figure 1. Study area & photographic survey location of Areas A, B, and C.

international student status, which were coded as categorical variables;

- b. Locations that the respondents frequented (two questions): Respondents were asked to mark six spots on

East Magnolia Avenue and six spots on West Magnolia Avenue that they visited or passed using an online map. Images of the locations along the streets were provided for reference;

- c. Overall walking time per week including nighttime walks asked with a five-point ordinal scale (two questions); and
- d. Overall safety perceptions of streets and perceptions of specific physical street settings shown in the photographs (six questions): the three areas were rated based on daytime and nighttime scenarios using a five-point Likert scale, followed by open-ended questions asking for explanations for each rating.

The open-ended questions and safety perception ratings for each photograph were analyzed and 784 keywords were retrieved. The sentiments of keywords (phrases) within sentences were also categorized as positive, neutral, or negative. Positive keywords included: very, good, well, enough, nice, fine, high, or easily. Negative words included: only, not enough, poor, uncomfortable, no, low, uneven, sketchy, difficult, worse, uneasy, bother, or little. Keywords that explained the status of the street setting such as “it is a setback,” “busy area,” or “heavy traffic” were also sorted. As advised by Liu (2012), the responses were carefully examined to detect any sarcasm accompanying positive words (e.g., “What a great car! It stopped working in two days”) and phrases that included relevant words but that did not express sentiment (e.g., “if I can find a good camera in the shop, I will buy it.”).

The Wilcoxon–Mann–Whitney test and Analysis of Variance (ANOVA) were conducted to identify the differences in walking time and safety perceptions by personal

characteristics. Spearman’s rank-order correlation was used to identify the association between walking time and the level of safety perceptions by using STATA 14.0. Then, content and sentiment analyses were performed. Each keyword from three open-ended questions that were asked about the perceptions of the three areas A, B, and C in the photographs was classified into one of three groups: (1) street setting, (2) personal feeling, and (3) general description. The “street setting” category was regrouped based on five elements that the authors used to measure each spot’s quality. The “personal characteristics and experience” category contained keywords describing the respondents’ characteristics (e.g., “Being a female,” “I am a man,” “I’m a small white female,” and “I concealed a handgun”) and certain contextual circumstances (e.g., “The cops are less than 30 seconds away,” “Strictly a residential area. No cause for any fear,” and “I walk alone”). The “general description” category included illustrations of areas but was not specifically related to street settings: for example, daytime, nighttime, bright, dark, nothing, hot, and normal.

4. Results

4.1. Descriptive statistics of respondents

Of the respondents, 83.1% (133 out of 160 people) were 19–25 years old and the rest were older than 25; in addition, 57.5% (92 out of 160) identified as female. Most respondents (91.9%) were domestic students; 86.9% identified as white and only 2.5% identified as African American. The Asian respondents identified as either Chinese, Thai, or Korean, accounting for 5.0% in total and 1.9% declined to report their race/ethnicity. Regarding time spent outdoors, 58.1% of respondents walked for around 30 minutes to an hour per week and 27.5% of respondents walked for 1–3 hours. 13.1% (21 people) walked for less than 30 minutes per week, and only 1.3% (two people) walked for more than 3 hours per week. Most respondents (89.4%) walked for less than 30 minutes per week after dark.

Overall, respondents felt safe in this study area based on the average score of 2.0 on the scale where 1.0 was the highest level of perceived safety and 5.0 was the lowest (Table 2). The gap between nighttime and daytime evaluations was largest for Area A: 3.3 at night and 1.35 during the day. Area B’s average perceived safety was similar, with nighttime and daytime scores of 2.1 and 1.4, respectively. The average safety perception scores shared a correlation with the area quality measured by the authors; for example, Area A was seen as the least safe place and also had the lowest quality street setting.

Table 1. Surveyed areas & their quality measures.

Quality measures		Area A	Area B	Area C
Imageability	Historic Buildings	0	2	0
	Identifiers	0	2	1
	Accessible Open Space	1	0	0
Human Scale	Street Furniture	0	2	1
	Lights on Buildings	0	2	1
Transparency	Long Sight Line	2	1	2
	First Floor Window	0	1	1
	Day & Night Activity	0	2	1
Complexity	Pedestrian Activity	0	2	2
	Business Type & Density	0	2	1
	Building Color & Design	0	2	1
	Outdoor Dining	0	1	1
	Public Art	0	1	0
Tidiness	Ped. Infrastructure	1	2	2
	Traffic Density & Speed	1	1	1
	Noise & Smells	2	0	1
	Sidewalk	2	2	2
	Litter & Graffiti	2	1	1
	Landscape & Vegetation	2	0	1
Average Score		0.68	1.37	1.05

2-point: Good, 1-point: Moderate, 0-point: Bad.

Table 2. Perception toward Areas A, B, and C of respondents.

	Area A		Area B		Area C	
	Night	Day	Night	Day	Night	Day
Very Safe (1-point)	11 (6.9%)	122 (76.3%)	60 (37.7%)	139 (87.4%)	30 (19.0%)	111 (69.8%)
Somewhat Safe (2-point)	28 (17.5%)	21 (13.1%)	54 (34.0%)	5 (3.1%)	41 (25.9%)	23 (14.5%)
Okay/Neutral (3-point)	46 (28.8%)	3 (1.9%)	26 (16.4%)	0 (0.0%)	52 (32.9%)	14 (8.8%)
Somewhat Unsafe (4-point)	52 (32.5%)	5 (3.1%)	14 (8.8%)	2 (1.3%)	27 (17.1%)	3 (1.9%)
Very Unsafe (5-point)	23 (14.4%)	3 (1.9%)	5 (3.1%)	13 (8.2%)	8 (5.1%)	8 (5.0%)
Total	160 (100%)	154 (100%)	159 (100%)	159 (100%)	158 (100%)	159 (100%)
Average Scores	3.30	1.35	2.06	1.40	2.62	1.58

4.2. Activity mapping

Respondents were asked to mark the spots that they frequented. The highest activity densities—the red color in Figures 2 and 3 indicates a higher level of activity while the blue color indicates a lower density—occurred in areas located within 0.25 miles (about 400 m) of a downtown intersection that has commercial businesses such as restaurants, retail shops, and bars. In addition, spots with well-equipped pedestrian infrastructure such as crosswalks, stop signs, and ground lights showed relatively high activity density because they connected pedestrian and vehicle campus entrances with nearby mixed-use areas. Area A, which is mostly residential and located further from the downtown area on East Magnolia Avenue, showed the lowest level of pedestrian activity.

4.3. Comparison by different groups

The results suggest that walking time varies statistically according to age; students under 25 years old tended to walk for longer periods each week ($z = -2.56, p = 0.01$), whereas other individual characteristics such as gender and international student status did not have an influence. Similarly, students under 25 years old tended to walk more during the nighttime ($z = -2.05, p = 0.04$). The results of ANOVA indicate that there is no statistical difference in walking time, including nighttime walking, among the different categories of race. Expressed nervousness did not vary statistically in terms of individual characteristics, except for gender. The Wilcoxon–Mann–Whitney test indicated that female

students generally felt unsafe on the streets ($z = -4.00, p = 0.00$). In terms of location, female students were more likely to feel unsafe in Area A ($z = -5.52, p = 0.00$) and Area C ($z = -4.11, p = 0.00$) at night. During the day, there was no statistical difference based on gender. Spearman's rank-order correlation reported that walking time and safety perception did not have any statistically significant association.

4.4. Content and sentiment analysis

Among the 784 keywords from the three open-ended questions, 214 were classified as general descriptions and 49 words were classified as personal characteristics and experience; the rest described the street settings.

The sentiments and keywords were identified for each photographed area (more detailed information can be found in Appendix). As shown in Figure 4, *lighting* was mentioned the most (139 times) followed by *pedestrian* (92 times), *bright* (89 times), *pedestrian infrastructure* (58 times), *daytime* (52 times), *familiarity with the place* (52 times), and *dark* (51 times). By drawing a comparison between day and night, *street lighting* was found to be a significant determinant of safety perception ratings at night, whereas *volume of pedestrians* functioned as a detriment during the day.

Only considering street settings, at Area A, which was perceived as somewhat unsafe by respondents, *street lighting*, *tree shadows*, *hidden spots*, and *pedestrians* were the most frequently mentioned factors. At Area B, which was perceived as safe, *street lighting*, *pedestrians*, and *number of business* were relevant factors. At Area C, which was



Figure 2. Activity density map: west side of Magnolia Avenue.



Figure 3. Activity density map: east side of Magnolia Avenue.



Figure 4. The word crowd of key words by frequency.

evaluated as being relatively neutral, *street lighting*, *pedestrian infrastructure*, *volume of pedestrians*, and *hidden spots* were the most highlighted. Although Areas A, B, and C were all perceived as safe during the day with scores of 1.35, 1.40, and 1.58 out of 5 points, respectively, the reasoning differed between them. *Large volume of pedestrians*, *high visibility*, and *pedestrian infrastructure* were mentioned most frequently for Area A. *Large volume of pedestrians*, *different businesses*, and *familiarity with the place* were mentioned the most for Area B. *Poor or good pedestrian infrastructure* and *good visibility* were highlighted the most for Area C; the same factor (e.g., *pedestrian infrastructure*) could be considered either good or poor by different respondents. Meanwhile, Area A was perceived as the least safe place during the nighttime with a score of 3.0. This was the opposite trend to the daytime score. *Poor lighting* was mentioned the most frequently for Area A. Meanwhile, good lighting was stated the most frequently for Areas B and C with respective nighttime scores of 2.06 and 2.62.

When categorizing the descriptions of respondents in street quality domains (see [Table 1](#) for detailed categorizations), the day scene at Area A was described positively with safety and tidiness (22 out of 128), complexity (17), and transparency (17) being mentioned the most frequently. At night were *human scale* (42 out of 183) and *transparency* (27), both of which were negatively expressed. At Area B, complexity was mentioned most frequently (44 out of 176 and 26 out of 114) followed by imageability (42 and 17), both at night and during the day. The safety and tidiness category was addressed least frequently in this area. For Area C's night scene, respondents expressed concerns about safety and tidiness (27) and human scale (10), whereas human scale (23) and complexity (17) were described as positive elements. Few comments were mentioned for the day scene in Area C. In summary, as shown in [Figure 5](#), respondents expressed significantly different sentiments according to the area, particularly for the night scene.

5. Discussion

A good urban space can enhance a community's social, physical, and psychological well-being (Mehta, 2007). In particular, streets comprise a substantial portion of the public realm. For example, streets and sidewalks constitute 24% of Chicago's land area and >70% of its public open space (Project for Public Spaces, 2015). These structures come to represent the city as a whole (Appleyard, Gerson & Lintell, 1981; Jacobs, 1961); for example, Fifth Avenue of Manhattan in New York City, Rodeo Drive in Los Angeles, and

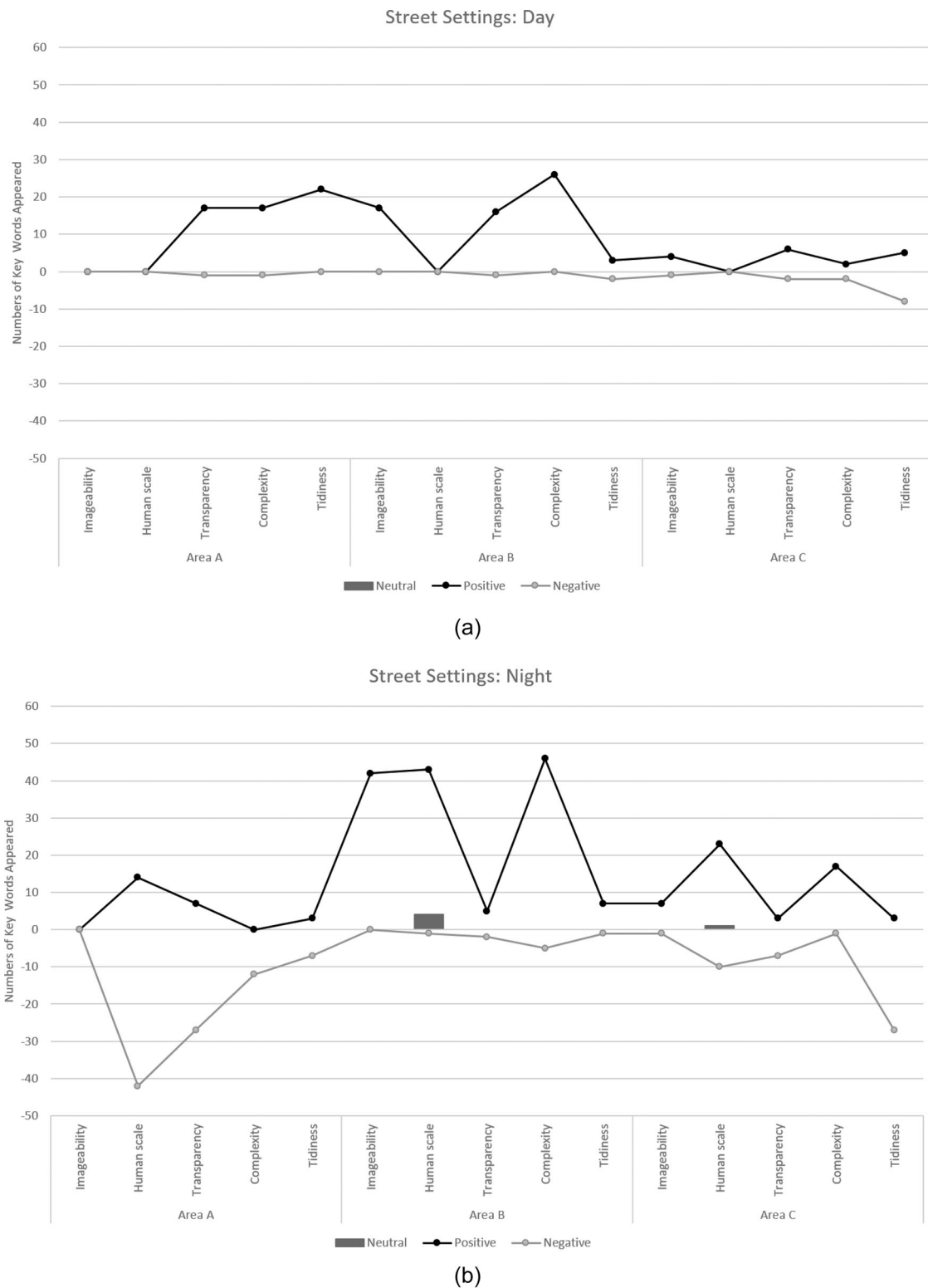


Figure 5. The positive and negative sentiments by area, street setting domains, and time.

Michigan Avenue in Chicago. Furthermore, streets provide the basic framework for a city, connecting and dividing areas and affecting how people move (Lee & Moudon, 2004; Park, Huang & Newman, 2016; Southworth & Owens, 1993). Billions of dollars have already been invested to promote better access for street travelers. The needs of non-motorized transportation users such as pedestrians, cyclists,

and transit riders should take priority in future investments toward streets.

This study's findings provide evidence to support these investments in streets since the perception and determinants of street settings because such perception strongly affects pedestrians' walking decisions. In extreme cases, fear of the imagined unknown can result in agoraphobic anxieties that



Figure 6. Area B, which found the most activity regardless less tidiness.

diminish the use of public spaces. This study offers several findings that are relevant to future street plans and designs. First, younger college students tend to walk more than older students (≥ 25 years old), although neither walking time nor night walking time differ by gender, race, or international status. This is because older students are more likely to live off campus, requiring them to use cars more often. Increasing the survey pool may cause these results to differ. However, the present outcomes seem promising because early-life exposure to and experience with regular walking may be the best way to develop lifelong active habits (Appleton, Holdsworth & Kubzansky, 2016; Shahid & Bertazzon, 2015). Thus, planning and design policies may consider targeting these young adults and listening to their needs to enhance their pleasant street experiences.

Second, the gender difference raises an important discussion. Female students felt less safe and comfortable and expressed strong levels of apprehension in areas with lower imageability, human scale, and complexity, even though the streets were tidy and clean. This finding aligns with those of previous studies that claim that females tend to experience more fear due to their vulnerability to offenders (Fetchenhauer & Buunk, 2005; Fisher & Nasar, 1992); these were also expressed in their responses. An open-ended question asked why the respondent gave a certain rating to a shown photograph; some respondents mentioned that the state of being female determines their level of perception toward the safety on the street.

"Being a female, being outside alone after dark is nerve-wracking."

"I'm a small white female."

"Because it is not well lit up at the end. Wouldn't feel safe as a female walking alone"

Several European cities including Vienna have adopted female-friendly plans that may be categorized as "gender mainstreaming." This planning strategy focuses on women's different needs, characteristics, perceptions, and transportation behaviors (Damyanovic, 2013). For example, women tend to walk and use public transit more than men, so

additional street lighting at night, widened sidewalks for strollers and children, and ramps next to staircases would enhance their experiences on streets (Damyanovic, 2013). The goal of female-friendly planning is to promote gender equity through diversity, participation, mobility, and accessibility. Indeed, female-friendly planning does not just apply to women; if a place is appropriate for women, this would benefit all people regardless of age, gender, or disability status. Moreover, analysis of public space usage based on different people groups can enhance the design of these public spaces. Future research may focus on how, why, when, and how long each group of people tends to use these public spaces.

Third, the responses to the open-ended questions specify the factors that determine people's safety perception toward certain street settings. Overall, darkness is strongly related to negative feelings, so proper street lighting is critical for reinforcing feelings of safety, which is consistent with previous findings. Interestingly, street lighting was seen as negative for homogeneous residential areas and positive for areas with mixed-use and better pedestrian facilities, even though illumination levels did not differ substantially and streetlights were installed at longer intervals in mixed-use areas. This partially demonstrates that streetlights cannot function as a panacea. Relatively high morphologic complexity in the form of busy corners stimulating day and night pedestrian activities and building fronts close to the streets would help mediate pedestrian fear. When some respondents described an area in Figure 6 with the highest activity density, they often mentioned incivilities related to "drunken" or "crazy" people.

"It is dark, but there are still lights so you can see what's around you and businesses/residences nearby. I also know that there are usually people around downtown you can call to for help."

"Toomer's corner is safe whether or not it's light on or dark outside, at least to me."

"It's dark but a popular area."

This outcome is consistent with the arguments of Jacobs (1961) and other researchers, who indicated that the presence of "natural watchers and guardians in sufficient numbers" due to mixed land use tends to encourage people to walk outside at all hours. A future study would benefit more by having additional open-ended questions to further explore the merits of qualitative research. A large data set and path analysis would help find causal connections between the levels of safety perceptions and street settings, personal characteristics, and other attributes (Peng, Feng & Timmermans, 2019).

Up-zoning a single-use district to mixed land use might be difficult. However, numerous theoretical and empirical studies have praised the benefits of mixed land use, so these projects require careful attention. Newly developed areas face few barriers to assigning diverse functions and activities. For redevelopments, zoning revision is essential. Up-zoning—and occasionally down-zoning—itself can be an issue, as zoning is closely related to property value and residents are sensitive to this. Additionally, any negative spillover

from newly allowed nonresidential uses could cause problems for existing residents in adjacent residential neighborhoods. Infill development within unoccupied areas could help increase the overall compactness and diversity of land use while not harming existing neighborhoods (Park, 2017). To lessen the negative spillover from these redevelopments, gradual development intensity, compatible functions and activities, and buffer zones (e.g., green strips, landscape, and vegetation) should be implemented. These landscape buffers are often examples of safety and tidiness, so their supplementary role in redevelopment is essential based on the results of the comments in our study. Neighborhood plans could support detailed design guides based on contextually unique neighborhood characteristics. Future research can further this cause by investigating people's perception of the types, proportions, and locations of optimal land use mixtures to mediate the fear and anxiety associated with streets.

People tend to emphasize slightly different street features based on the time of day. As indicated, street lighting is a major concern at night, followed by pedestrians' familiarity, sidewalks and pedestrian infrastructure, hidden spots and shadows, and nearby businesses. During the day, people express more concerns about openness, sidewalks, and pedestrian infrastructure, nearby businesses, and familiarity. For example, trees were described in opposite ways based on the time of the day.

"The trees and grass are pleasant to look at." and "It is green and populated."

vs. "The large tree by the sidewalk creates a large blind spot", "a little dark because of the tree shade," and "dark lighting/mass shadows from big trees."

Trees—which are often considered street amenities—are perceived positively during the day and negatively at night because the hidden spots they can create tend to diminish pedestrian visibility. There are two ways to interpret this outcome. First, design guides may attempt to create convenient, comfortable, and safe streets and planners should balance the two-fold demands of nighttime and daytime use. Specific guidelines may be provided; for instance, the police department for the city of Boise provides the following landscaping guides for crime prevention: "If planting within six feet of a sidewalk, ..., these plants should not exceed a height of two feet at maturity or be trimmed to this height." (Boise Police, 2018). Donovan and Prestemon (2012) studied 2,813 homes in Portland from 2005 to 2007 to investigate the relationship between trees and crime; they found that neighborhoods with lined trees and larger yard trees experienced fewer property crimes; similar results were found in other studies (Gilstad-Hayden & Meyer, 2015; Troy, Morgan Grove & O'Neil-Dunne, 2012). Thus, government policies and urban design strategies may attempt to reduce the gaps between crime and the fear of crime (Beatty, Grimsley, Lawless, Manning, & Wilson, 2005). Future studies should investigate which and how street settings are perceived differently according to the results of local crime reports.

Lastly, this study demonstrates the potential use of advanced online survey tools such as Qualtrics in urban design procedures. Qualtrics can create heat maps and

hotspot maps while asking respondents to click specific spots for different interests. Using the Google Maps API service, researchers and policymaker may immediately geocode addresses associated with particular questions. For example, they may ask survey respondents to name the place they love the most within a city. Then, the researchers can easily generate a density or hotspot map. New survey tools like this could help professionals in planning and urban design provide place-specific solutions more easily and quickly.

The findings of this study could bring some meaningful implementation points in urban planning and design but there would be a high chance that respondents were influenced by their personal experience and previous knowledge about a place. As indicated, most respondents have lived in the city for a while and are familiar with the surroundings, which can lead to the underestimation of some valuation elements in safety perceptions; familiarity with a place was mentioned 52 times in open-ended questions. Future research should adopt different models to investigate safety perceptions and built environments such as observational methods or choice modeling approaches. Behavior observation methods highlighted the observation of naturally occurring behavioral patterns of pedestrians that can possibly reduce survey respondents' preexisting knowledge and experiences. As Bian, Ma, Rong, Wang, and Lu (2009) suggested, survey and observation methods can complement each other by mediating investigators' and surveyors' biases. Sample selection bias may arise from repeated visitors like among the respondents to this study. Similar to Apostolakis and Jaffry (2006), two stages of models could be adopted for both first-time and repeat visitors. In addition, choice modeling with a series of discrete choice sets would benefit the detection of changes of marginal value in each attribute instead of asking the rank order of preference toward one attribute (Hanley, Mourato & Wright, 2002).

6. Conclusion

This research found interesting results that reinforce the previous literature. The professionals' and survey respondents' evaluations of street settings were similar, although nonprofessionals did not use technical terms. Since gender affects safety perception and walking activities, the concept of gender mainstreaming should be taken into account with regard to street design. In the same condition, people's perceptions vary based on their backgrounds and personal characteristics. Predicting these traits is difficult, but street settings could help reduce the fear of streets regardless. The outcome of this study is that the complexity arising from different types of land use, businesses, and pedestrian streams on the street plays a more significant role than distinctive identifiers in places, high levels of tidiness, and incivilities in determining the level of pedestrian safety perception. In addition, this study opens a discussion about the use of online tools to collect location-specific data and effectively implement the data in planning and urban design processions. This would provide better access for feedback on street improvements and developing street design guides.

Funding

This work was supported by Auburn University, USA under the Undergraduate Research Fellowship Program of 2017–2018.

ORCID

Yunmi Park  <http://orcid.org/0000-0002-7924-6518>

References

- Apostolakis, A., & Jaffry, S. (2006). Correcting for sample selection bias in stated preference tourist surveys. *Tourism Economics*, 12(3), 451–468. doi:10.5367/000000006778493600
- Appleton, A. A., Holdsworth, E. A., & Kubzansky, L. D. (2016). A systematic review of the interplay between social determinants and environmental exposures for early-life outcomes. *Current Environmental Health Reports*, 3, (3), 287–301. doi:10.1007/s40572-016-0099-7
- Appleton, J. (1975). *The experience of place*. London: Wiley.
- Appleyard, D., Gerson, M. S., & Lintell, M. (1981). *Livable streets, protected neighborhoods*. Berkeley, CA: University of California Press.
- Beatty, C., Grimsley, M., Lawless, P., Manning, J., & Wilson, I. (2005). *Fear of crime in NDC areas: How do perceptions relate to reality*. Sheffield: Centre for Regional Economic and Social Research, Sheffield Hallam University.
- Bereitschaft, B. (2017). Equity in microscale urban design and walkability: A photographic survey of six Pittsburgh streetscapes. *Sustainability*, 9(7), 1233. doi:10.3390/su9071233
- Bian, Y., Ma, J., Rong, J., Wang, W., & Lu, J. (2009). Pedestrians' level of service at signalized intersections in China. *Transportation Research Record: Journal of the Transportation Research Board*, 2114(1), 83–89. doi:10.3141/2114-10
- Boise Police. (2018). Landscaping for crime prevention. Boise.
- Brownson, R. C., Baker, E. A., Housemann, R. A., Brennan, L. K., & Bacak, S. J. (2001). Environmental and policy determinants of physical activity in the United States. *American Journal of Public Health*, 91, 1995–2003. doi:10.2105/AJPH.91.12.1995
- Chiricos, T., Hogan, M., & Gertz, M. (1997). Racial composition of neighborhood and fear of crime. *Criminology*, 35(1), 107–132. doi:10.1111/j.1745-9125.1997.tb00872.x
- Cohen, D. A., McKenzie, T. L., Sehgal, A., Williamson, S., Golinelli, D., & Lurie, N. (2007). Contribution of public parks to physical activity. *American Journal of Public Health*, 97(3), 509–514. doi:10.2105/AJPH.2005.072447
- Cozens, P. M., Saville, G., & Hillier, D. (2005). Crime prevention through environmental design (CPTED): A review and modern bibliography. *Property Management*, 23(5), 328–356. doi:10.1108/02637470510631483
- Damyanovic, D. (2013). Gender mainstreaming as a strategy for sustainable urban planning. In *Fair shared cities. The impact of gender planning in Europe* (pp. 177–192). London: Routledge.
- Donovan, G. H., & Prestemon, J. P. (2012). The effect of trees on crime in Portland, Oregon. *Environment and Behavior*, 44(1), 3–30. doi:10.1177/0013916510383238
- Duany, A., Plater-Zyberk, E., Krieger, A., & Lennertz, W. R. (1991). *Towns and town-making principles*. New York: Rizzoli International Publications.
- Ewing, R., & Clemente, O. (2013). *Measuring urban design: Metrics for livable places*. Washington, DC: Island Press.
- Fanghanel, A. (2015). "The trouble with safety: Subjectification, Safety and of Fear of Crime in Public Space." *Theoretical Criminology*, 20(1), 57–74.
- Ferraro, K. F., & Grange, R. L. (1987). The measurement of fear of crime. *Sociological Inquiry*, 57(1), 70–97. doi:10.1111/j.1475-682X.1987.tb01181.x
- Fetchenhauer, D., & Buunk, B. P. (2005). How to explain gender differences in fear of crime: Towards an evolutionary approach. *Sexualities, Evolution & Gender*, 7, 95–113. doi:10.1080/00207170500111044
- Fisher, B. S. (1995). Crime and fear on campus. *The Annals of the American Academy of Political and Social Science*, 539(1), 85–101. doi:10.1177/0002716295539001007
- Fisher, B. S., & Nasar, J. L. (1992). Fear of crime in relation to three exterior site features: Prospect, refuge and escape. *Environment and Behavior*, 24(1), 35–65. doi:10.1177/0013916592241002
- Fox, K. A., Nobles, M. R., & Piquero, A. R. (2009). Gender, crime victimization and fear of crime. *Security Journal*, 22(1), 24–39. doi:10.1057/sj.2008.13
- Frumkin, H. (2002). Urban sprawl and public health. *Public Health Reports (Washington, D.C. : 1974)*, 117(3), 201–217. doi:10.1093/phr/117.3.201
- Gibson, C. L., Zhao, J., Lovrich, N. P., & Gaffney, M. J. (2002). Social integration, individual perceptions of collective efficacy, and fear of crime in three cities. *Justice Quarterly*, 19(3), 537–564. doi:10.1080/07418820200095341
- Gilstad-Hayden, K., & Meyer, S. R. (2015). Trees, a new partner in the fight against urban crime. US App-American Politics and Policy Blog.
- Grant, J. (2002). Mixed use in theory and practice: Canadian experience with implementing a planning principle. *Journal of the American Planning Association*, 68(1), 71–84. doi:10.1080/01944360208977192
- Hale, C. (1996). Fear of crime: A review of the literature. *International Review of Victimology*, 4(2), 79–150. doi:10.1177/026975809600400201
- Hanley, N., Mourato, S., & Wright, R. E. (2002). Choice modelling approaches: A superior alternative for environmental valuation?. *Journal of Economic Surveys*, 15(3), 435–462. doi:10.1111/1467-6419.00145
- Hunter, A., & Baumer, T. L. (1982). Street traffic, social integration, and fear of crime. *Sociological Inquiry*, 52(2), 122–131. doi:10.1111/j.1475-682X.1982.tb01244.x
- Jacobs, J. (1961). *The death and life of great American cities*. New York: Vintage.
- Lee, C., & Moudon, A. V. (2004). Physical activity and environment research in the health field: Implications for urban and transportation planning practice and research. *Journal of Planning Literature*, 19(2), 147–181. doi:10.1177/0885412204267680
- Lee, J. S., Park, S., & Jung, S. (2016). Effect of crime prevention through environmental design (CPTED) measures on active living and fear of crime. *Sustainability*, 8(9), 872. doi:10.3390/su8090872
- Liu, B. (2012). Sentiment analysis and opinion mining. *Synthesis Lectures on Human Language Technologies*, 5(1), 1–167. doi:10.2200/S00416ED1V01Y201204HLT016
- Maxfield, Michael, G. (1984). The limits of vulnerability in explaining fear of crime: A comparative neighborhood analysis. *Journal of research in Crime and Delinquency*, 21(3), 233–250.
- Mehta, V. (2007). Lively streets: Determining environmental characteristics to support social behavior. *Journal of Planning Education and Research*, 27(2), 165–187. doi:10.1177/0739456X07037947
- Minnery, J. R., & Lim, B. (2005). Measuring crime prevention through environmental design. *Journal of Architectural and Planning Research*, 22, 330–341.
- Montgomery, J. (1998). Making a city: Urbanity, vitality and urban design. *Journal of Urban Design*, 3, (1), 93–116. doi:10.1080/13574809808724418
- Montoya, L., Junger, M., & Ongena, Y. (2016). The relation between residential property and its surroundings and day-and night-time residential burglary. *Environment and Behavior*, 48(4), 515–549. doi:10.1177/0013916514551047
- Nicks, D. (2016). Why There's Been a Huge Decline in Drivers' Licenses for Millennials and Gen X, Time.
- Ortega, S. T., & Myles, J. L. (1987). Race and gender effects on fear of crime: An interactive model with age. *Criminology*, 25(1), 133–152. doi:10.1111/j.1745-9125.1987.tb00792.x
- Painter, K. (1996). The influence of street lighting improvements on crime, fear and pedestrian street use, after dark. *Landscape and Urban Planning*, 35(2–3), 193–201. doi:10.1016/0169-2046(96)00311-8

- Park, Y. (2015). The network of patterns: Creating a design guide using Christopher Alexander's pattern language. *Environment and Planning B: Planning and Design*, 42(4), 593–613. doi:10.1068/b130011p
- Park, Y. (2017). Does new urbanist neighborhood design affect neighborhood turnover?. *Land Use Policy*, 68, 552–562. doi:10.1016/j.landusepol.2017.07.013
- Park, Y., Huang, S.-K., & Newman, G. D. (2016). A statistical meta-analysis of the design components of new urbanism on housing prices. *Journal of Planning Literature*, 31(4), 435. 0885412216667898. doi:10.1177/0885412216667898
- Peng, Y., Feng, T., & Timmermans, H. (2019). A path analysis of outdoor comfort in urban public spaces. *Building and Environment*, 148, 459–467. doi:10.1016/j.buildenv.2018.11.023
- Perkins, D. D., Meeks, J. W., & Taylor, R. B. (1992). The physical environment of street blocks and resident perceptions of crime and disorder: Implications for theory and measurement. *Journal of Environmental Psychology*, 12(1), 21–34. doi:10.1016/S0272-4944(05)80294-4
- Poyner, B., & Webb, B. (1997). Reducing theft from shopping bags in city centre markets. In: R. V. Clarke (Ed.) *Situational crime prevention: Successful case studies*. Guilderland, NY: Harrow and Heston.
- Project for Public Spaces. (2015). *Streets as Places Toolkit*, Public Space Resources, New York, NY.
- Ratnayake, R. (2016). Association between fear of crime gender, student nationality, and physical features. *Environment and Behavior*, 1, 1–16.
- Shahid, R., & Bertazzon, S. (2015). Local spatial analysis and dynamic simulation of childhood obesity and neighbourhood walkability in a major Canadian city. *AIMS Public Health*, 2(4), 616–637. doi:10.3934/publichealth.2015.4.616
- Sivak, M., & Schoettle, B. (2016). Recent decreases in the proportion of persons with a driver's license across all age groups.
- Skogan, W. G., & Maxfield, M. G. (1981). *Coping with crime: Individual and neighborhood differences*. Beverly Hills: Sage.
- Southworth, M., & Owens, P. M. (1993). The evolving metropolis: Studies of community, neighborhood, and street form at the urban edge. *Journal of the American Planning Association*, 59(3), 271–287. doi:10.1080/01944369308975880
- Talen, E. (1999). Sense of community and neighbourhood form: An assessment of the social doctrine of new urbanism. *Urban Studies*, 36(8), 1361–1379. doi:10.1080/0042098993033
- Taylor, J., Eitle, D., & Russell, D. (2009). Racial/ethnic variation in the relationship between physical limitation and fear of crime: An examination of mediating and moderating factors. *Deviant Behavior*, 30(2), 144–174. doi:10.1080/01639620802050213
- Troy, A., Morgan Grove, J., & O'Neil-Dunne, J. (2012). The relationship between tree canopy and crime rates across an urban–rural gradient in the greater Baltimore region. *Landscape and Urban Planning*, 106(3), 262–270. doi:10.1016/j.landurbplan.2012.03.010

Appendix: Frequency of key words from survey by areas

Area A: Night		Area A: Day		Area B: Night		Area B: Day		Area C: Night		Area C: Day	
Lighting–	42	Bright	46	Lighting+	43	Bright	24	Lighting+	23	Bright	19
Dark	36	Daytime	23	Familiarity	36	Daytime	16	Ped. Infra–	21	Daytime	13
Lighting+	14	Pedestrian↑	17	Pedestrian↑	30	Business↑	13	Pedestrian↑	17	Ped. Infra–	7
Tree shadow–	12	Visibility+	14	Business↑	14	Pedestrian↑	13	Lighting–	10	Ped. Infra+	5
Belief of safety+	12	Ped. Infra+	10	Dark	7	Familiarity	11	Dark	8	Visibility+	5
Feeling–	11	Landscape+	6	Lighting-neutral	4	Open	7	Hidden spot	7	Public space	2
Hidden spot	7	Clean	5	Public space	4	Visibility+	6	Driving–	5	Familiarity	2
Pedestrian↓	6	Open	3	Pedestrian↓	5	Belief of safety+	5	Close to campus	4	Feeling+	3
Business↓	6	Belief of safety+	1	Info. of crime	4	Public space	3	Familiarity	3	Driving–	1
Visibility+	5	Security+	1	Security+	6	Ped. Infra+	3	No hidden spot	3	Tree shadow–	1
Gender-female	4	Isolated	1	Belief of safety+	3	Wide	2	Belief of safety+	2	Belief of safety+	1
Empty space–	4	Pedestrian↓	1	Gender-female	2	Center of town	2	Walking alone	2	Business↑	1
Walking alone	4			No hidden spot	2	Info. of crime	2	Ped. Infra+	2	Building design–	1
Visibility–	4			Hidden spot	2	No Hidden spot	1	Feeling–	2	Wide	1
Ped. Infra–	4			Center of town	2	Hidden spot	1	Lighting -neutral	1	Visibility–	1
Ped. Infra+	3			Wide	2	Historical place	1	Driving+	1	Unfamiliarity	1
Self-defense	3			Business↑	2	Hot	1	Pedestrian↓	1	Pedestrian↑	1
Parked cars–	2			Visibility+	1	Ped. Infra–	1	Normal	1	Pedestrian↓	1
Landscape+	1			Night time	1	Normal	1	Security–	1		
Open	1			Self-defense	1	Driving–	1	Info. of crime	1		
Info. of crime	1			Lighting–	1			Unfamiliarity	1		
Tree shadow+	1			Nothing	1			Self-defense	1		
				Feeling–	1						
				Ped. Infra–	1						
				Ped. Infra+	1						
183		128		176		114		117		66	

+refers to as positive,

–as negative,

↓as low, and

↑as high.